

Toon Racer Starter Track Kit

Technical Documentation & User Guide



Created by: Apex Studio

Version: 1.0.0

Engine: Unity **Render Pipeline:** Built-in

--- 1. INTRODUCTION ---

Welcome to the **Toon Racer: Starter Track Kit** by Apex Studio!

Thank you for downloading this package. This kit is designed specifically for game developers looking for high-quality, ready-to-use stylized (toon) racing environments and vehicles. We have optimized these assets to ensure seamless integration into your project with absolutely no complex coding required.

1.1. Pack Overview

This lightweight, drag-and-drop package includes the essential building blocks to prototype or visually enhance your stylized racing game:

- **5 Unique Racing Tracks:** Each track is built as a single, continuous mesh for optimal performance, smooth lighting, and immediate scene placement.
- **1 Stylized Hero Car:** A high-quality toon-style vehicle mesh, perfectly scaled and ready to be attached to your preferred vehicle physics controller.
- **1 Modular Barrier:** A versatile, standalone curb piece designed for detailing custom track layouts.

1.2. Key Features

- **Drag-and-Drop Workflow:** All prefabs are pre-configured. Simply drag them from the Project window into your Hierarchy or Scene view and start creating.
- **Performance Optimized:** Utilizing single continuous meshes for the main tracks significantly reduces draw calls, making it highly efficient for both mobile and PC platforms.
- **Clean Geometry:** Professional-grade topology designed to work flawlessly with stylized shading and dynamic lighting.

--- 2. FOLDER STRUCTURE ---

To maintain a clean, professional, and scalable project environment, all assets are strictly organized under the Apex Studio root directory.

- **Apex Studio/**
 - **Toon Racer Starter Track Kit/**
 - **Demo/**
 - **Scenes/** (Contains the demo scene showcasing all assets and track layouts)
 - **Documentation/** (Contains offline guides and reference materials)
 - **Materials/** (Pre-configured materials applied to the models)
 - **Meshes/** (Raw FBX models for the tracks, vehicle, and props)
 - **Prefabs/** (Ready-to-use, fully configured prefabs for drag-and-drop workflow)
 - **Tracks/** (Folder containing the 5 unique continuous track prefabs)
 - **Barrier** (The modular curb/barrier prefab)
 - **Car** (The stylized hero car prefab)
 - **Ground** (The base ground prefab)
 - **Textures/** (High-quality color palettes and texture maps)

--- 3. QUICK START GUIDE ---

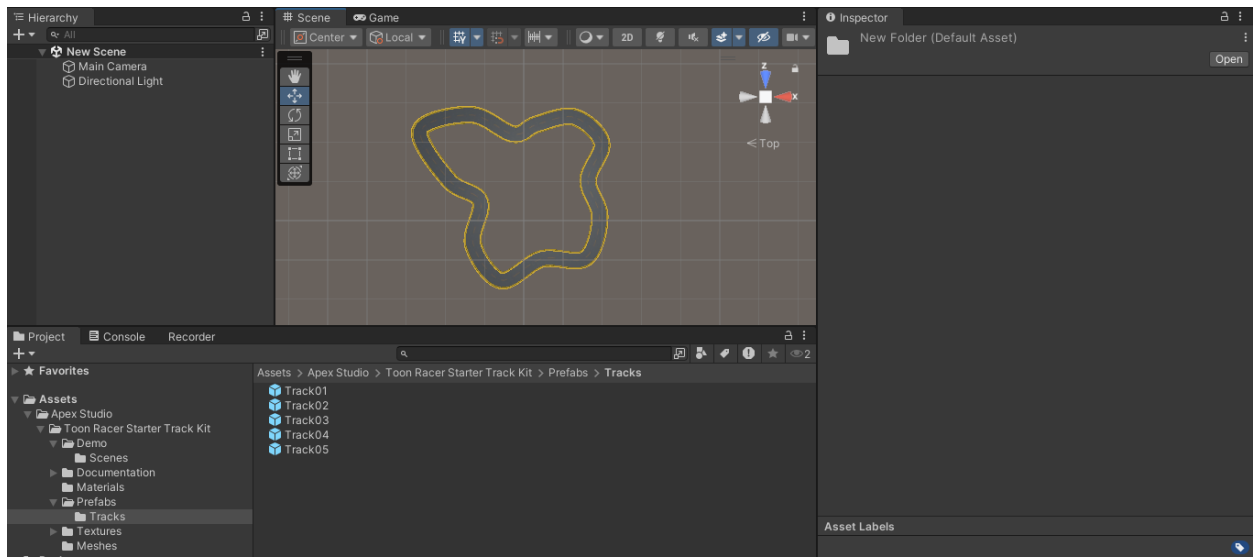
Integrating the Stylized Racing Asset Pack into your game is incredibly straightforward. No coding or complex setup is required.

3.1. Importing the Package

1. Open your Unity Project.
2. Navigate to **Assets > Import Package > Custom Package...**
3. Locate the downloaded .unitypackage file and select it.
4. In the Import window, ensure all files are checked and click **Import**.

3.2. Adding Assets to Your Scene

1. In your Project window, navigate to **Apex Studio / Toon Racer Starter Track Kit /Prefabs/**.
2. Drag and drop any prefab (Track, Car, or Barrier) directly into your Scene view or Hierarchy.
3. **Pro Tip:** Always reset the Transform of the instantiated track prefab (Position 0, 0, 0) to ensure perfect alignment if you are snapping modular pieces like the curbs.



--- 4. TECHNICAL SPECIFICATIONS ---

At Apex Studio, we prioritize performance without compromising on visual quality. The **Toon Racer Starter Track Kit** is engineered to be highly optimized for all platforms, including mobile, standalone PC, and consoles.

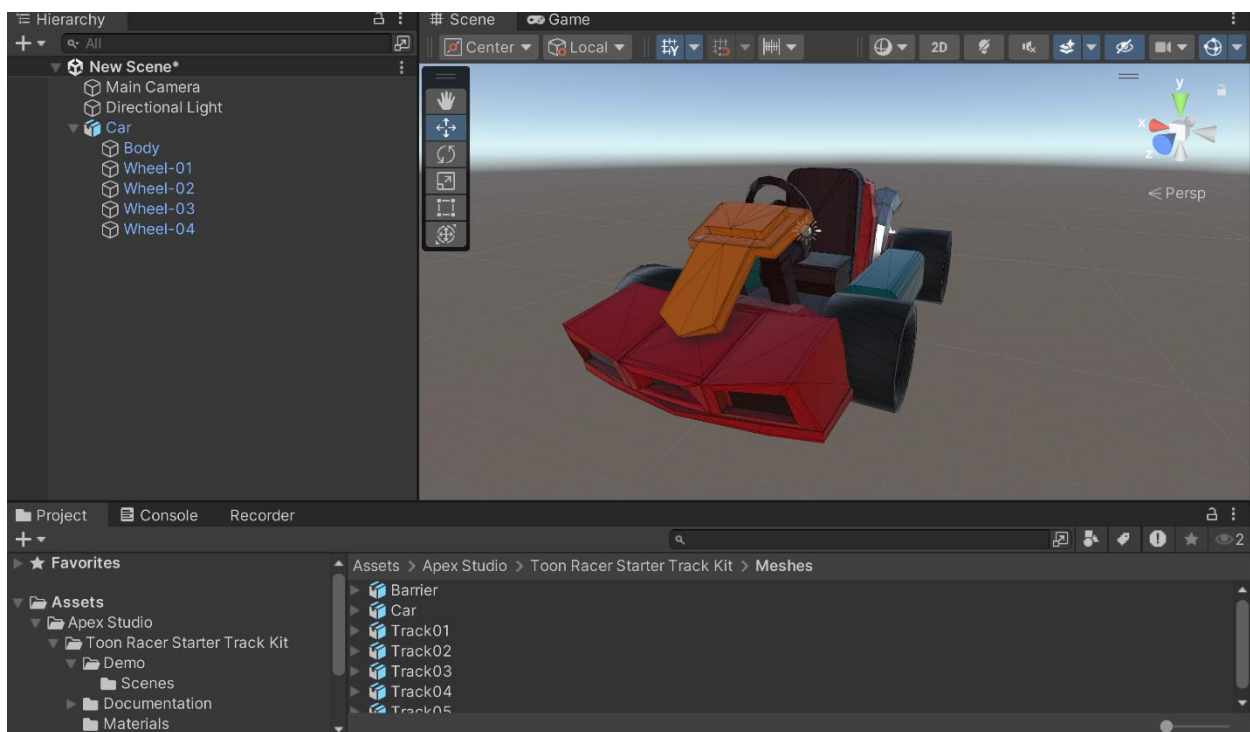
4.1. Mesh & Topology Optimization

- **Continuous Track Meshes:** Each of the 5 tracks is constructed as a single, continuous mesh. This deliberate design choice drastically reduces draw calls and batching overhead, ensuring maximum frame rates even on lower-end devices.
- **Clean Geometry:** All models feature professional-grade, clean topology (quads and optimized triangles). There are no overlapping vertices or unnecessary hidden faces.



4.2. Asset Breakdown

- **Tracks (x5):** Distinct layouts (e.g., straightaways, sharp curves, elevation changes). Scaled accurately to Unity's standard metric system (1 unit = 1 meter).
- **Stylized Hero Car (x1):** The vehicle mesh is cleanly separated (Main Body and 4 individual Wheels) with correctly centered pivot points. This makes it instantly compatible with any standard vehicle physics controller (such as Unity's Wheel Colliders) or arcade-style hover scripts.
- **Barrier & Ground (x2):** Modular design. The barrier pivot points are specifically placed at the base corners to allow for seamless vertex snapping along the track edges.



4.3. Materials & Texturing

- **Shared Materials:** To further enhance performance, the assets share a unified, stylized color palette/texture atlas. This ensures minimal material swapping during rendering.
- **UV Mapping:** Non-overlapping, cleanly unwrapped UVs optimized for the stylized shading pipeline.

4.4. Collision Guidelines

To keep the package lightweight and flexible, we have applied standard colliders:

- **Tracks:** Default Mesh Colliders are applied for precise surface driving.
- **Props/Car:** Simple Box/Convex colliders are applied. *(Note: You may adjust or replace these with your custom physics rig requirements).*

Section 5: Technical Specifications & Asset Scaling

5.1 Track Layout Architecture

To ensure optimal hardware performance, seamless physics calculation, and zero mesh-alignment gaps, all racing tracks in this kit have been designed and exported from Blender as Single-Mesh Layouts rather than separated modular pieces. This production-ready approach minimizes the draw-call count, making the package highly optimized for mobile and standalone platforms.

5.2 Engine Scale & Units Consistency

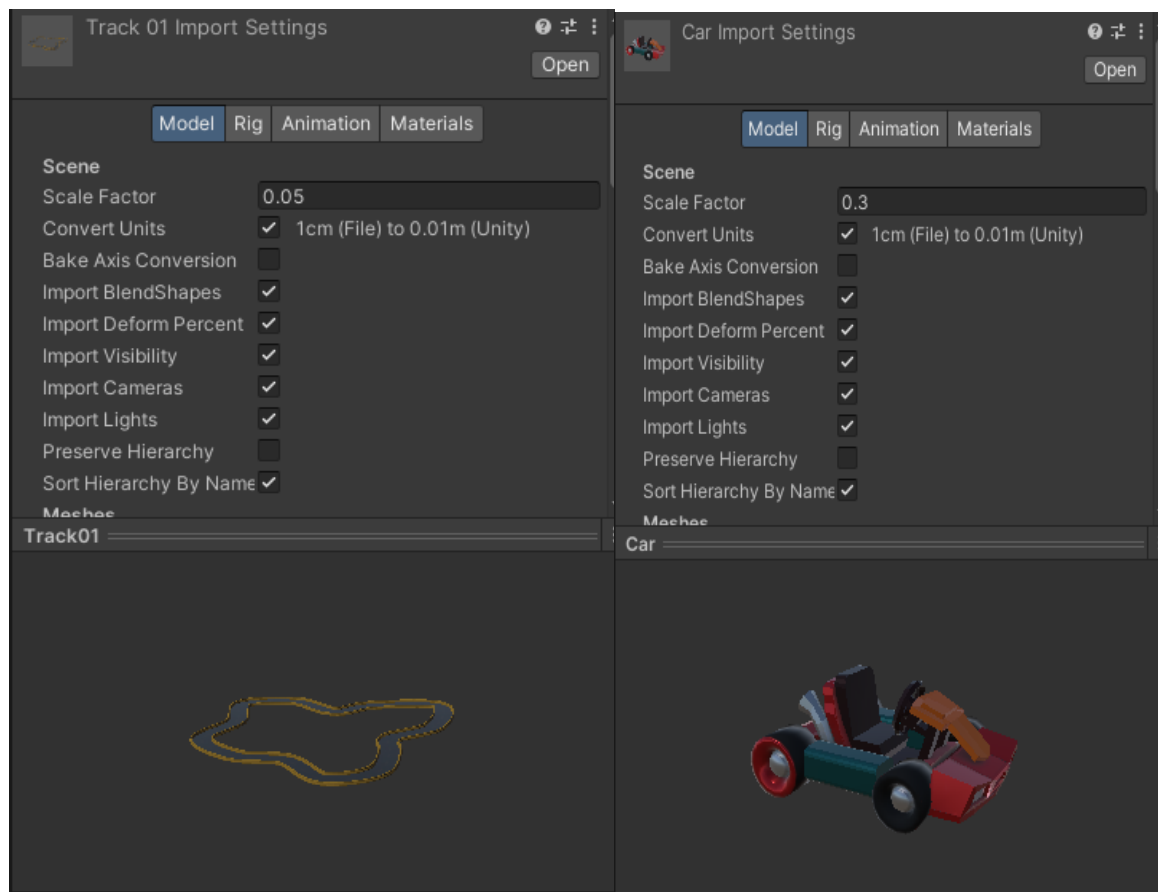
All assets (Tracks and Vehicles) have been meticulously calibrated to match Unity's default coordinate and measurement system (1 Unit = 1 Meter).

- **Global Scale Factor:** The tracks are pre-scaled via Unity's Model Import Settings (Scale Factor) to seamlessly align with standard physics and the default Unity Cube ($1\text{m} \times 1\text{m} \times 1\text{m}$).
- **Vehicle Scale:** The vehicle dimensions are perfectly proportioned relative to the track width, ensuring a realistic, arcade-ready gameplay experience right out of the box.

5.3 Custom Scaling & Adjustments

If your project requirements utilize a custom-scaled controller or distinct gravity metrics, you can scale the assets without losing any mesh fidelity.

- **How to Adjust:** Select the desired track or vehicle model in the Project Window, navigate to the Inspector, and modify the Scale Factor under the *Model* tab (as shown in the illustration below), then click Apply. Alternatively, you can directly scale the ready-to-use Prefabs provided in the package.



--- 6. SUPPORT & COMMUNITY ---

Thank you once again for choosing the **Toon Racer Starter Track Kit** by Apex Studio. We are committed to providing high-quality tools for developers and ensuring our assets meet the highest industry standards.

5.1. Need Assistance?

If you encounter any technical issues, have questions regarding implementation, or would like to request specific features for future updates, please do not hesitate to reach out to us.

- **Support Email:** [apexforgehq@gmail.com]
- **ArtStation / Portfolio:** [<https://www.artstation.com/apex-studio>]

5.2. Feedback & Reviews

Your feedback is invaluable to us! If this asset pack has streamlined your game development workflow or added value to your project, please consider taking a moment to leave a review on the Unity Asset Store. Honest ratings help us grow, improve, and continue creating high-quality stylized assets for the developer community.

Happy Developing! - *The Apex Studio Team*